
Dyadic Inverse Germs and Barnes–Rvachev Deconvolution

New frontier results for the inverse Fabius function,
Kabaya–Iri polynomials, Thue–Morse differences, exact
 q -Richardson recovery, reciprocal-sinc poles, and Lambert W

A research synthesis based on the complete recursive LaTeX corpus in
Analysis/FabiusFunction/docs of Vladimir Reshetnikov’s
ProveIt repository

Research status. The report distinguishes three levels of provenance. Statements marked REPO are established in the audited repository; statements marked PRIOR are imported from the external literature, notably the scaling-polynomial work of S. L. Lee; and statements marked NEW are proved here and were not located explicitly in the 57-path repository corpus by path, phrase, formula, and construction searches. “New” is corpus-relative and is not a claim of worldwide priority. The external search found the Kabaya–Iri/Appell precursor, so the novelty claims below are deliberately restricted to the dyadic inverse-germ, Barnes, Thue–Morse, exact-recovery, logistic-duality, pole-asymptotic, and coefficient-threshold deductions.

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Abstract

The Fabius–Rvachev system has two natural inverse problems. The first is nonlinear: invert the Fabius distribution function near a point. The second is linear: invert averaging against the Rvachev probability law on polynomials. The repository develops the ingredients of both problems, but not the common algebra that emerges when they are placed side by side.

For every interior dyadic point $x_0 = a/2^N$, the Taylor series of F terminates. We show that the Taylor germ of $G = F^{-1}$ at $F(x_0)$ is therefore the convergent algebraic inverse of a finite rational polynomial. The true inverse differs from this algebraic branch by a nonzero flat function. At $x_0 = 1/4$ the algebraic shadow is a Catalan square-root branch, giving the exact all-order formula

$$G^{(m)}(5/72) = m!(-1)^{m-1}4^{m-1}C_{m-1}.$$

This converts exact dyadic Fabius arithmetic into an explicit, Bell-polynomial and Lagrange–Bürmann calculus for the inverse.

For the linear problem, we begin from the known Kabaya–Iri scaling polynomials and recenter them at the symmetric Rvachev law. Their reciprocal moment generating function yields a centered Appell sequence A_n . We derive a Bernoulli-cumulant and Bell-polynomial calculus, and exhibit a new dual random variable: an infinite dyadic sum of logistic variables whose characteristic function is the reciprocal Rvachev moment generating function. Consequently $A_n(x) = \mathbb{E}(x + iZ)^n$, giving exact sign alternation of the Appell constants.

Finite dyadic convolution prefixes are identified exactly with normalized Barnes multiple Bernoulli polynomials at the period vector $(1, 1/2, \dots, 2^{1-N})$. Applying the complete Thue–Morse difference operator to these polynomials collapses them to one monomial, with a closed factor $2^{-N(N-1)/2}n^{\underline{N}}$. Tail self-similarity then shows that every finite-prefix polynomial is an *exact* polynomial in 2^{-N} (or 4^{-N} after centering). Gaussian q -binomial Lagrange weights therefore recover the infinite Kabaya–Iri polynomial exactly after finitely many prefix levels; this is Richardson extrapolation without a remainder.

Finally, the reciprocal moment generating function is meromorphic with poles at $2\pi in$ of order $\nu_2(n) + 1$. We transfer the full 2-adic zero divisor of the infinite sinc product into large-degree asymptotics of the Appell constants. The first three pole shells give an explicit three-term expansion with a polynomial correction from the double pole at $4\pi i$. Stirling inversion then produces a second, coefficient-theoretic role for the Lambert W function: the first index at which the Appell constants exceed a prescribed size is $\log X/(2W(\log X/(2\pi e))) + O(1)$.

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1 Scope, corpus audit, and result boundary

1.1 What was audited

The target tree is not a single manuscript. It contains a primary exposition, a mathematical glossary, a Lean walkthrough, a consolidated frontier volume, an active Thue–Morse formula atlas, twelve Fourier-decay dossiers, seven source-paper transcriptions, and thirty-two frozen historical sources. The recursive ledger contains 57 distinct `.tex` paths: 25 living paths and 32 archive paths. The complete path list is reproduced in [appendix A](#).

The overlap audit used four complementary tests.

1. Current successors were read at source level, including the main Fabius–Rvachev exposition, the frontier volume, the formula atlas, the inverse supplement, the q -special-function exposition, the Fourier-decay synthesis, and the latest interpolation/extrapolation reports.
2. Frozen versions were treated as prior art, not discarded as obsolete. Revision chains and archive provenance maps were used to identify claims unique to earlier drafts.
3. Exact phrase and formula searches were run for Appell, Sheffer, umbral, Barnes–Bernoulli, multiple Bernoulli, mean-value inversion, polynomial deconvolution, and the new pole and inverse-germ formulas.
4. A targeted literature search was made after the repository audit. It found Lee’s 2009 construction of the Kabaya–Iri biorthogonal Appell polynomials for the up-function. The report therefore imports that construction rather than relabeling it as new.

Nonduplication boundary

The repository already contains extensive work on exact dyadic Fabius values, the global signed extension, terminating Taylor series, the totalized inverse, inverse endpoint asymptotics, Gaussian q -binomial interpolation, Thue–Morse Prouhet cancellation, finite splines, sharp convolution errors, Bernoulli cumulants, Bell-polynomial saddle expansions, logarithmic sinc-product Richardson extrapolation, phase-locked Lambert nodes, Fourier zero divisors, arithmetic rays, and a higher-rank Pascal–Rvachev hierarchy. None of those results is reproved as a purported novelty. The new contribution is the set of compositions among them listed in [table 1](#).

1.2 Result map

1.3 Why the two inverse problems belong together

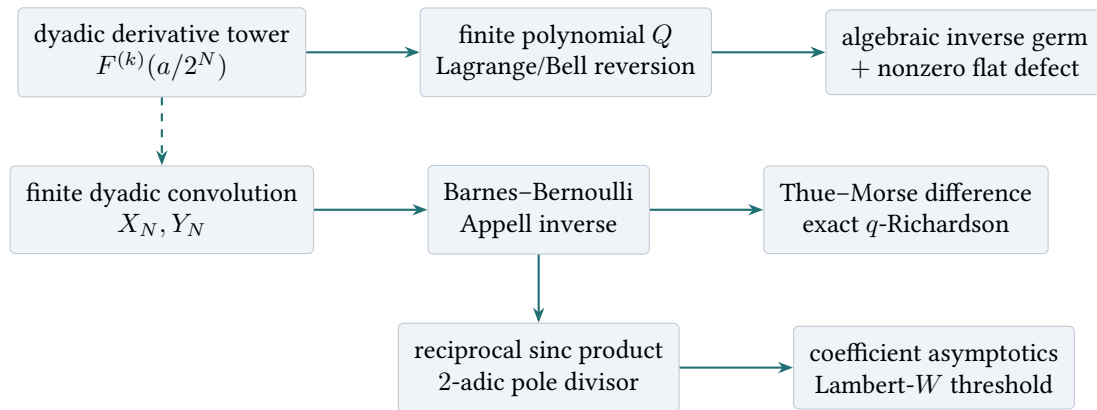
Let μ be the probability law of the Fabius random variable X , and let ν be the centered Rvachev law of $Y = 2X - 1$. There are two different inverses:

$$\begin{aligned} \text{nonlinear inverse:} \quad & F(x) = y \mapsto x = G(y), \\ \text{linear polynomial inverse:} \quad & (\mathcal{T}_\nu p)(x) := \int p(x + y) \nu(dy) \mapsto \mathcal{T}_\nu^{-1} x^n = A_n(x). \end{aligned}$$

The first is governed by compositional inversion and therefore by Lagrange–Bürmann and partial Bell polynomials. The second is governed by reciprocal exponential generating functions and therefore by Appell, Barnes, Bernoulli, and complete Bell polynomials. Dyadic self-similarity makes both inverse calculi finite at special scales: the nonlinear Taylor jet terminates at a dyadic point, and the finite convolution prefix is a finite Barnes product.

Object	Principal statement	Status
Dyadic inverse germ	The Taylor series of G at $F(a/2^N)$ is the algebraic inverse of a finite rational polynomial; the discrepancy from G is nonzero and flat.	NEW
Catalan point	$G^{(m)}(5/72) = m!(-1)^{m-1}4^{m-1}C_{m-1}$.	NEW
Kabaya–Iri sequence	$e^{xt}/L(t) = \sum P_n(x)t^n/n!$ and biorthogonality to derivatives of the up/Fabius density.	PRIOR
Centered Appell sequence	$A_n(x) = 2^n P_n((x+1)/2)$, with Bernoulli cumulants and Bell constants.	derived
Logistic dual	$1/M(t)$ is the characteristic function of a dyadic logistic series, so $A_n(x) = \mathbb{E}(x + iZ)^n$.	NEW
Dyadic Barnes prefix	Finite-prefix polynomials are normalized Barnes multiple Bernoulli polynomials at a geometric period vector.	NEW
Thue–Morse operator	A complete binary signed difference of a finite-prefix Appell polynomial collapses exactly to a monomial.	NEW
Exact q -Richardson recovery	The limiting Appell polynomial is recovered exactly from finitely many convolution-prefix levels by Gaussian q -binomial weights.	NEW
Reciprocal-sinc poles	Pole order $\nu_2(n) + 1$ becomes a degree- $\nu_2(n)$ polynomial correction in the large-degree Appell constants.	NEW
Lambert threshold	The first coefficient above size X has index $\log X/(2W(\log X/(2\pi e))) + O(1)$.	NEW

Table 1: The result boundary used throughout the report.



2 Established platform and notation

2.1 Fabius, the signed extension, and the Rvachev density

Let $F : \mathbb{R} \rightarrow [0, 1]$ be the bounded Fabius function: $F = 0$ on $(-\infty, 0]$, $F = 1$ on $[1, \infty)$, and on the unit interval

$$F(1-x) = 1 - F(x), \quad F'(x) = 2F(2x) \quad (0 \leq x \leq 1/2). \quad (1)$$

The Rvachev up-function is the even density supported on $[-1, 1]$ for which

$$F'(x) = 2 \operatorname{up}(2x - 1), \quad 0 < x < 1. \quad (2)$$

The repository's signed global extension $\tilde{F}$ agrees with F on $[0, 1]$ and satisfies

$$\tilde{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \tilde{F}(2^m x). \quad (3)$$

At nonnegative integers,

$$\tilde{F}(2r) = 0, \quad \tilde{F}(2r+1) = \varepsilon_r, \quad \varepsilon_r = (-1)^{\operatorname{wt}_2(r)}. \quad (4)$$

Consequently, if $x_0 = a/2^N$ is a reduced dyadic point with a odd, then

$$F^{(m)}(x_0) = 0 \quad (m > N). \quad (5)$$

All derivatives in the finite jet are rational, because the scaled arguments in (3) are dyadic and the repository gives exact rational evaluators there.

2.2 The two random series

Let $(U_j)_{j \geq 1}$ be independent uniform random variables on $[0, 1]$, and let $(V_j)_{j \geq 1}$ be independent uniform random variables on $[-1, 1]$. Put

$$X = \sum_{j \geq 1} 2^{-j} U_j, \quad Y = \sum_{j \geq 1} 2^{-j} V_j = 2X - 1. \quad (6)$$

Then F is the distribution function of X and u is the density of Y . Their moment generating functions are entire:

$$L(t) = \mathbb{E} e^{tX} = \prod_{j \geq 1} \frac{e^{t/2^j} - 1}{t/2^j}, \quad (7)$$

$$M(t) = \mathbb{E} e^{tY} = \prod_{j \geq 1} \frac{\sinh(t/2^j)}{t/2^j} = e^{-t} L(2t). \quad (8)$$

Self-similarity gives

$$L(2t) = \frac{e^t - 1}{t} L(t), \quad M(2t) = \frac{\sinh t}{t} M(t). \quad (9)$$

On the imaginary axis,

$$M(2\pi iz) = \prod_{k \geq 0} \operatorname{sinc}(z/2^k) =: \Phi(z), \quad \operatorname{sinc} u = \frac{\sin(\pi u)}{\pi u}. \quad (10)$$

Thus the moment problem and the infinite sinc product are two restrictions of the same entire function.

2.3 Moments and cumulants

Write $\kappa_n(X)$ and $\kappa_n(Y)$ for cumulants. The Bernoulli expansion of $\log((e^t - 1)/t)$ gives

$$\kappa_1(X) = \frac{1}{2}, \quad \kappa_{2r}(X) = \frac{B_{2r}}{2r(4^r - 1)}, \quad \kappa_{2r+1}(X) = 0 \quad (r \geq 1), \quad (11)$$

$$\kappa_1(Y) = 0, \quad \kappa_{2r}(Y) = \frac{2^{2r-1} B_{2r}}{r(4^r - 1)}, \quad \kappa_{2r+1}(Y) = 0. \quad (12)$$

The first centered moments are

$$m_0 = 1, \quad m_2 = \frac{1}{9}, \quad m_4 = \frac{19}{675}, \quad m_6 = \frac{583}{59535}, \quad m_8 = \frac{132809}{32531625}. \quad (13)$$

These established rational sequences will become coefficients in the exact finite-prefix expansion.

3 Algebraic Taylor shadows of the inverse Fabius function

3.1 The finite dyadic jet polynomial

Fix a reduced dyadic point

$$x_0 = \frac{a}{2^N} \in (0, 1), \quad a \text{ odd}, \quad y_0 = F(x_0). \quad (14)$$

Put

$$f_k = F^{(k)}(x_0) = 2^{\binom{k+1}{2}} \tilde{F}(2^k x_0), \quad Q_{x_0}(h) = \sum_{k=1}^N \frac{f_k}{k!} h^k. \quad (15)$$

Since $F'(x_0) > 0$, the polynomial Q_{x_0} has a nonzero linear term. The smooth Taylor remainder

$$\rho_{x_0}(h) := F(x_0 + h) - y_0 - Q_{x_0}(h) \quad (16)$$

is flat at $h = 0$: every derivative of ρ_{x_0} vanishes there. It is not identically zero on any neighborhood, because F is nowhere analytic in $(0, 1)$.

Theorem 3.1 (Dyadic algebraic inverse germ; NEW). *Let $G = F^{-1} : (0, 1) \rightarrow (0, 1)$. There is a unique analytic germ S_{x_0} at 0 such that*

$$Q_{x_0}(S_{x_0}(z)) = z, \quad S_{x_0}(0) = 0. \quad (17)$$

It is an algebraic function over $\mathbb{Q}(z)$. Moreover

$$G(y_0 + z) = x_0 + S_{x_0}(z) + R_{x_0}(z), \quad (18)$$

where R_{x_0} is C^∞ , flat at 0, and nonzero in every neighborhood of 0. Consequently, the complete Taylor series of G at y_0 is convergent and algebraic even though it does not represent G on any neighborhood.

Proof. The analytic inverse-function theorem applied to the polynomial Q_{x_0} gives the unique analytic germ S_{x_0} . Since S_{x_0} satisfies the polynomial equation $Q_{x_0}(w) - z = 0$, it is algebraic over $\mathbb{Q}(z)$; rationality of the coefficients of Q_{x_0} follows from the exact dyadic evaluator.

Let

$$H(z) = G(y_0 + z) - x_0.$$

Then

$$z = F(x_0 + H(z)) - y_0 = Q_{x_0}(H(z)) + \rho_{x_0}(H(z)).$$

The composite $\rho_{x_0} \circ H$ is flat because $H(0) = 0$ and $H'(0) > 0$. Comparing derivatives recursively in this equation shows that the formal Taylor series of H is the unique formal inverse of Q_{x_0} , hence equals the Taylor series of S_{x_0} . Thus $R_{x_0} = H - S_{x_0}$ is flat.

If R_{x_0} vanished on a neighborhood, then G would agree there with the analytic function $x_0 + S_{x_0}$. Since $G'(y_0) > 0$, the analytic inverse-function theorem would make F analytic at x_0 , contradicting the nowhere-analyticity theorem. Therefore R_{x_0} is nonzero in every neighborhood. $\square$

The paradox in its sharpest form

At a dyadic output $y_0 = F(a/2^N)$, the inverse Fabius function has a Taylor series with a positive radius of convergence, and that series is an algebraic function with rational coefficients. Yet the sum of the series is not the inverse Fabius function on either side of y_0 : the discrepancy is a nontrivial flat germ invisible to every derivative.

3.2 Lagrange–Bürmann and Bell formulas

Factor

$$Q_{x_0}(u) = u\varphi_{x_0}(u), \quad \varphi_{x_0}(0) = f_1 > 0. \quad (19)$$

The coefficients of the algebraic inverse are completely explicit.

Theorem 3.2 (All inverse derivatives at dyadic outputs; NEW). *For every $m \geq 1$,*

$$G^{(m)}(y_0) = (m-1)! [u^{m-1}] \varphi_{x_0}(u)^{-m}. \quad (20)$$

Equivalently, if $g_m = G^{(m)}(y_0)$ and $B_{m,k}$ denotes the partial exponential Bell polynomial, then

$$g_1 = \frac{1}{f_1}, \quad (21)$$

$$g_m = -\frac{1}{f_1} \sum_{k=2}^{\min(m,N)} f_k B_{m,k}(g_1, g_2, \dots, g_{m-k+1}) \quad (m \geq 2). \quad (22)$$

In particular, every $G^{(m)}(y_0)$ is rational.

Proof. The Lagrange–Bürmann formula for the inverse of $z = u\varphi_{x_0}(u)$ gives

$$[z^m] S_{x_0}(z) = \frac{1}{m} [u^{m-1}] \varphi_{x_0}(u)^{-m}.$$

Multiplication by $m!$ proves (20). For the Bell recurrence, differentiate $F(G(y)) = y$ m times at $y = y_0$. Faà di Bruno gives

$$\sum_{k=1}^m f_k B_{m,k}(g_1, \dots, g_{m-k+1}) = 0 \quad (m \geq 2).$$

Since $B_{m,1} = g_m$, solving for the newest derivative gives (22). All inputs are rational. $\square$

Remark 3.3. The formulas use two different Bell-polynomial roles in the same theory. Partial Bell polynomials invert composition in (22); complete Bell polynomials will invert a moment generating function in section 4. The two appearances are not cosmetic variants: one organizes set partitions by the number of blocks, while the other sums over all block counts.

3.3 The midpoint: a linear Taylor shadow

At $x_0 = 1/2$ one has $y_0 = 1/2$, $F'(1/2) = 2$, and all higher derivatives vanish. Hence

$$Q_{1/2}(h) = 2h, \quad S_{1/2}(z) = \frac{z}{2}. \quad (23)$$

Thus

$$G(1/2 + z) = \frac{1}{2} + \frac{z}{2} + R_{1/2}(z), \quad (24)$$

where $R_{1/2}$ is a nonzero flat odd germ. Every derivative of G at $1/2$ beyond the first is zero, but G is not linear on any neighborhood.

3.4 The quarter point: Catalan numbers exactly

The first nontrivial case is unexpectedly rigid.

Theorem 3.4 (Catalan inverse jet at $F(1/4)$; NEW). *The exact value and derivative jet at $x_0 = 1/4$ are*

$$F(1/4) = \frac{5}{72}, \quad F'(1/4) = 1, \quad F''(1/4) = 8, \quad F^{(m)}(1/4) = 0 \quad (m \geq 3). \quad (25)$$

Therefore

$$Q_{1/4}(h) = h + 4h^2, \quad S_{1/4}(z) = \frac{\sqrt{1 + 16z} - 1}{8}. \quad (26)$$

For every $m \geq 1$,

$$G^{(m)}(5/72) = m!(-1)^{m-1}4^{m-1}C_{m-1}, \quad C_r = \frac{1}{r+1} \binom{2r}{r}. \quad (27)$$

The corresponding large-order law is

$$\left| G^{(m)}(5/72) \right| \sim \frac{m! 16^{m-1}}{\sqrt{\pi} (m-1)^{3/2}}. \quad (28)$$

Proof. The value $F(1/4) = 5/72$ is part of the repository's exact dyadic table. From (3),

$$F'(1/4) = 2F(1/2) = 1, \quad F''(1/4) = 8\tilde{F}(1) = 8,$$

and (5) removes all higher derivatives. Solving $z = h + 4h^2$ for the branch with $h(0) = 0$ gives (26). The classical Catalan generating function

$$\frac{\sqrt{1 + 16z} - 1}{8} = \sum_{m \geq 1} (-1)^{m-1} 4^{m-1} C_{m-1} z^m$$

proves (27). Stirling's estimate $C_r \sim 4^r / (\sqrt{\pi} r^{3/2})$ proves (28). $\square$

Example 3.5 (The first derivatives).

$$\begin{aligned} G'(5/72) &= 1, & G''(5/72) &= -8, & G^{(3)}(5/72) &= 192, \\ G^{(4)}(5/72) &= -7680, & G^{(5)}(5/72) &= 430080. \end{aligned}$$

These rapidly growing derivatives belong to the algebraic square-root shadow. They do not imply analyticity of G : the actual function differs by a flat germ.

3.5 The eighth point: a cubic algebraic shadow

At $x_0 = 1/8$, the exact value is $F(1/8) = 1/288$ and

$$F'(1/8) = \frac{5}{36}, \quad F''(1/8) = 4, \quad F^{(3)}(1/8) = 64. \quad (29)$$

Thus the inverse Taylor series at $1/288$ is the branch at the origin of

$$z = \frac{5}{36}h + 2h^2 + \frac{32}{3}h^3. \quad (30)$$

This gives an explicit cubic algebraic function and, through (20), a one-line coefficient formula

$$G^{(m)}(1/288) = (m-1)! [u^{m-1}] \left(\frac{5}{36} + 2u + \frac{32}{3}u^2 \right)^{-m}. \quad (31)$$

The quarter point is therefore not an isolated curiosity: every dyadic input produces a finite algebraic shadow, with degree equal to the reduced denominator exponent.

3.6 A general singularity program for inverse derivatives

The nearest singularities of S_{x_0} occur among the critical values

$$z_c = Q_{x_0}(u_c), \quad Q'_{x_0}(u_c) = 0. \quad (32)$$

When the nearest critical point is simple, S_{x_0} has a square-root singularity and its Taylor coefficients have the universal $m^{-3/2}$ factor seen in (28). Multiple critical points lead to other Puiseux exponents. Since Q_{x_0} is exactly computable over $\mathbb{Q}$, this converts the high-order inverse-derivative problem at every dyadic output into finite algebraic geometry.

Frontier problem: dyadic inverse singularity atlas

For each reduced dyadic $a/2^N$, classify the critical values of $Q_{a/2^N}$, determine which one sets the convergence radius of the inverse Taylor shadow, and describe the Galois and 2-adic structure of the resulting Puiseux constants. The case $N = 2$ is Catalan; the case $N = 3$ is cubic; already $N = 4$ should reveal whether a stable binary pattern exists.

4 Kabaya–Iri polynomials and the Rvachev Appell inverse

4.1 The external precursor and the normalization used here

Lee’s scaling-polynomial construction associates to a compactly supported probability density ϕ the Appell sequence generated by $e^{xt}/\hat{\phi}(it)$ and proves its biorthogonality to the distributional derivatives of ϕ . For the up-function, these are the Kabaya–Iri polynomials. In the Fabius normalization used in this report, it is convenient to distinguish the uncentered law $X \in [0, 1]$ from the centered law $Y = 2X - 1 \in [-1, 1]$.

Definition 4.1 (Kabaya–Iri and centered Rvachev–Appell polynomials). Define $P_n, A_n \in \mathbb{Q}[x]$ by

$$\frac{e^{xt}}{L(t)} = \sum_{n \geq 0} P_n(x) \frac{t^n}{n!}, \quad (33)$$

$$\frac{e^{xt}}{M(t)} = \sum_{n \geq 0} A_n(x) \frac{t^n}{n!}. \quad (34)$$

The affine change $M(t) = e^{-t}L(2t)$ gives

$$A_n(x) = 2^n P_n\left(\frac{x+1}{2}\right), \quad P_n(x) = 2^{-n} A_n(2x-1). \quad (35)$$

Thus the sequence P_n is the known Kabaya–Iri sequence, while A_n is its symmetric Rvachev normalization. The centering eliminates every odd cumulant and makes the pole analysis in [section 7](#) transparent.

4.2 Appell, mean-value, and biorthogonality identities

Proposition 4.2 (Classical Appell package; PRIOR). For $n \geq 1$,

$$P'_n(x) = nP_{n-1}(x), \quad A'_n(x) = nA_{n-1}(x), \quad (36)$$

$$P_n(x+y) = \sum_{k=0}^n \binom{n}{k} P_k(x) y^{n-k}, \quad A_n(x+y) = \sum_{k=0}^n \binom{n}{k} A_k(x) y^{n-k}. \quad (37)$$

The polynomial averaging operators are inverted exactly:

$$\boxed{\mathbb{E} P_n(x + X) = x^n, \quad \mathbb{E} A_n(x + Y) = x^n.} \quad (38)$$

If $f_X = F'$ is the density of X , then in the distributional sense

$$\left\langle (-1)^r f_X^{(r)}, \frac{P_n}{n!} \right\rangle = \delta_{rn}. \quad (39)$$

Equivalently, for the centered density up ,

$$\int_{-1}^1 (-1)^r \text{up}^{(r)}(x) \frac{A_n(x)}{n!} dx = \delta_{rn}. \quad (40)$$

Proof. Differentiate the generating functions with respect to x to obtain (36); multiply by e^{yt} to obtain (37). For the mean-value identities,

$$\mathbb{E} \sum_{n \geq 0} P_n(x + X) \frac{t^n}{n!} = \mathbb{E} \frac{e^{(x+X)t}}{L(t)} = e^{xt},$$

and similarly for A_n . The biorthogonality is Lee's general compactly supported distribution identity specialized to f_X and transported affinely to up . It can also be checked directly by repeated integration by parts and (38). $\square$

Remark 4.3. Equation (38) is an exact polynomial deconvolution formula. The operator $\mathcal{T}_Y : p \mapsto \mathbb{E} p(\cdot + Y)$ is triangular on $\mathbb{Q}[x]$ with ones on its diagonal; $A_n = \mathcal{T}_Y^{-1} x^n$. This is the linear inverse problem announced in section 1.

4.3 Bell reconstruction from Bernoulli cumulants

Set

$$\alpha_n = A_n(0), \quad \frac{1}{M(t)} = \sum_{n \geq 0} \alpha_n \frac{t^n}{n!}. \quad (41)$$

Symmetry gives $\alpha_{2r+1} = 0$. If $\mathcal{B}_n$ denotes the complete exponential Bell polynomial, then (12) yields

$$\boxed{\alpha_n = \mathcal{B}_n(-\kappa_1(Y), -\kappa_2(Y), \dots, -\kappa_n(Y)).} \quad (42)$$

Equivalently,

$$\alpha_0 = 1, \quad \alpha_n = - \sum_{j=1}^n \binom{n-1}{j-1} \kappa_j(Y) \alpha_{n-j}. \quad (43)$$

The first constants are

$$\alpha_0 = 1, \quad \alpha_2 = -\frac{1}{9}, \quad \alpha_4 = \frac{31}{675}, \quad \alpha_6 = -\frac{2347}{59535}, \quad \alpha_8 = \frac{1904369}{32531625}. \quad (44)$$

The Appell translation law then gives

$$A_n(x) = \sum_{k=0}^n \binom{n}{k} \alpha_{n-k} x^k. \quad (45)$$

For reference,

$$\begin{aligned}
 A_0(x) &= 1, \\
 A_1(x) &= x, \\
 A_2(x) &= x^2 - \frac{1}{9}, \\
 A_3(x) &= x^3 - \frac{1}{3}x, \\
 A_4(x) &= x^4 - \frac{2}{3}x^2 + \frac{31}{675}, \\
 A_5(x) &= x^5 - \frac{10}{9}x^3 + \frac{31}{135}x, \\
 A_6(x) &= x^6 - \frac{5}{3}x^4 + \frac{31}{45}x^2 - \frac{2347}{59535}.
 \end{aligned} \tag{46}$$

Using (35), the first uncentered polynomials are

$$\begin{aligned}
 P_0(x) &= 1, \\
 P_1(x) &= x - \frac{1}{2}, \\
 P_2(x) &= x^2 - x + \frac{2}{9}, \\
 P_3(x) &= x^3 - \frac{3}{2}x^2 + \frac{2}{3}x - \frac{1}{12}, \\
 P_4(x) &= x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{16}{675}, \\
 P_5(x) &= x^5 - \frac{5}{2}x^4 + \frac{20}{9}x^3 - \frac{5}{6}x^2 + \frac{16}{135}x - \frac{1}{270}.
 \end{aligned} \tag{47}$$

These agree with Lee’s displayed Kabaya–Iri list through degree five.

4.4 Bernoulli refinement recurrences

The self-similarity identities (9) become polynomial recurrences when the reciprocal generating functions are compared.

Proposition 4.4 (Bernoulli refinement of the Appell sequences). *Let B_k be defined by $t/(e^t - 1) = \sum B_k t^k/k!$, and let $c_{2r} = 2(1 - 2^{2r-1})B_{2r}$ be the exponential coefficients of $t/\sinh t$. Then*

$$2^n P_n(x) = \sum_{k=0}^n \binom{n}{k} B_k P_{n-k}(2x), \tag{48}$$

$$2^n A_n(x) = \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} c_{2r} A_{n-2r}(2x). \tag{49}$$

Proof. Substitute (9) into the defining generating functions:

$$\begin{aligned}
 \frac{e^{2xt}}{L(2t)} &= \frac{t}{e^t - 1} \frac{e^{2xt}}{L(t)}, \\
 \frac{e^{2xt}}{M(2t)} &= \frac{t}{\sinh t} \frac{e^{2xt}}{M(t)}.
 \end{aligned}$$

Coefficient comparison gives the two formulas. □

Remark 4.5. The repository’s Bernoulli recurrences usually govern moments, reciprocal sine products, or saddle coefficients. Here the same Bernoulli numbers govern a polynomial eigenbasis for the adjoint Rvachev averaging operator. The recurrence is a direct algebraic shadow of dyadic refinement.

4.5 A dyadic logistic dual

The complete Bell signs in (44) admit a probabilistic explanation that appears not to have been recorded for the Rvachev law.

Let Λ be a standard logistic random variable with density

$$\ell(x) = \frac{e^{-x}}{(1 + e^{-x})^2}. \quad (50)$$

The beta integral gives

$$\mathbb{E}e^{it\Lambda} = \Gamma(1 + it)\Gamma(1 - it) = \frac{\pi t}{\sinh(\pi t)}. \quad (51)$$

Let $(\Lambda_j)_{j \geq 1}$ be independent copies and define

$$Z = \sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j}. \quad (52)$$

The series converges in L^2 and almost surely.

Theorem 4.6 (Logistic dual of the Rvachev law; NEW). *For real t ,*

$$\mathbb{E}e^{itZ} = \prod_{j \geq 1} \frac{t/2^j}{\sinh(t/2^j)} = \frac{1}{M(t)}. \quad (53)$$

Consequently, as polynomial identities,

$$A_n(x) = \mathbb{E}(x + iZ)^n, \quad (54)$$

$$P_n(x) = \mathbb{E}\left(x - \frac{1}{2} + \frac{iZ}{2}\right)^n. \quad (55)$$

In particular,

$$\alpha_{2m} = (-1)^m \mathbb{E}Z^{2m}, \quad (-1)^m \alpha_{2m} > 0. \quad (56)$$

Proof. Independence and (51) give

$$\mathbb{E}e^{itZ} = \prod_{j \geq 1} \frac{t/2^j}{\sinh(t/2^j)} = \frac{1}{M(t)}.$$

The product converges locally uniformly on the real axis because the logarithm of its j th factor is $\mathcal{O}(4^{-j}t^2)$. Multiplying by e^{xt} and comparing the Taylor expansion at 0 proves (54). The affine identity (35) gives (55). Setting $x = 0$ and using symmetry of Z proves (56). $\square$

A duality of compact and noncompact laws

The Rvachev variable Y is compactly supported and has moment generating function M . Its polynomial deconvolution sequence is represented by imaginary moments of the noncompact variable Z , an infinite dyadic sum of logistic variables, whose characteristic function is M^{-1} . Thus the reciprocal sinc product is not merely a formal series: on the real axis it is itself a positive-definite characteristic function.

4.6 Operator interpretation

Let $D = d/dx$. On polynomials,

$$\mathcal{T}_Y = \mathbb{E}e^{YD} = M(D), \quad \mathcal{T}_Y^{-1} = M(D)^{-1} = \mathbb{E}e^{iZD}. \quad (57)$$

Therefore

$$A_n = M(D)^{-1}x^n. \quad (58)$$

This identity packages the mean-value theorem, the logistic dual, and the Appell derivative rule in one line. It also prepares the finite Barnes and Thue–Morse difference calculus: replacing M by a finite product replaces the infinite-order differential operator by a finite product of Bernoulli operators.

5 Finite dyadic prefixes as Barnes–Bernoulli polynomials

5.1 Finite random sums and finite Appell inverses

For $N \geq 1$, define the convolution prefixes

$$X_N = \sum_{j=1}^N 2^{-j}U_j, \quad Y_N = \sum_{j=1}^N 2^{-j}V_j, \quad (59)$$

with moment generating functions

$$L_N(t) = \prod_{j=1}^N \frac{e^{t/2^j} - 1}{t/2^j}, \quad (60)$$

$$M_N(t) = \prod_{j=1}^N \frac{\sinh(t/2^j)}{t/2^j}. \quad (61)$$

Define finite-prefix Appell polynomials by

$$\frac{e^{xt}}{L_N(t)} = \sum_{n \geq 0} P_n^{[N]}(x) \frac{t^n}{n!}, \quad (62)$$

$$\frac{e^{xt}}{M_N(t)} = \sum_{n \geq 0} A_n^{[N]}(x) \frac{t^n}{n!}. \quad (63)$$

They satisfy the exact finite mean-value identities

$$\mathbb{E}P_n^{[N]}(x + X_N) = x^n, \quad \mathbb{E}A_n^{[N]}(x + Y_N) = x^n. \quad (64)$$

Because X_N and Y_N are finite convolutions of boxes, these polynomials are the exact polynomial duals of the spline approximants developed elsewhere in the repository.

5.2 Normalized Barnes multiple Bernoulli polynomials

For nonzero periods $b_1, \dots, b_N$, define the normalized Barnes multiple Bernoulli polynomials $\mathfrak{B}_n^{(N)}$ by

$$e^{xt} \prod_{j=1}^N \frac{b_j t}{e^{b_j t} - 1} = \sum_{n \geq 0} \mathfrak{B}_n^{(N)}(x \mid b_1, \dots, b_N) \frac{t^n}{n!}. \quad (65)$$

This normalization has leading coefficient one and reduces to the ordinary Bernoulli polynomial when $N = 1$ and $b_1 = 1$.

Theorem 5.1 (Dyadic Barnes identification; NEW). *For every $N, n \geq 0$,*

$$P_n^{[N]}(x) = \mathfrak{B}_n^{(N)}(x \mid 2^{-1}, 2^{-2}, \dots, 2^{-N}). \quad (66)$$

Let

$$s_N = 1 - 2^{-N}, \quad (a_1, \dots, a_N) = (1, 1/2, \dots, 2^{1-N}). \quad (67)$$

Then

$$A_n^{[N]}(x) = \mathfrak{B}_n^{(N)}\left(x + s_N \mid 1, \frac{1}{2}, \dots, 2^{1-N}\right). \quad (68)$$

Proof. Equation (66) is immediate by comparing (62) with (65). For the centered product, write

$$\frac{\sinh(a_j t/2)}{a_j t/2} = e^{-a_j t/2} \frac{e^{a_j t} - 1}{a_j t}.$$

Since $\frac{1}{2} \sum_{j=1}^N a_j = s_N$, one has

$$\frac{e^{xt}}{M_N(t)} = e^{(x+s_N)t} \prod_{j=1}^N \frac{a_j t}{e^{a_j t} - 1},$$

which is (68). □

Remark 5.2. The period vectors are geometric rather than repeated. This is a Barnes analogue of the repository's ubiquitous $q = 1/2$ geometry: each new convolution scale appends one period to the multiple Bernoulli polynomial. The infinite Kabaya–Iri sequence is therefore a geometric-period limit of a Barnes tower.

5.3 A complete Thue–Morse difference operator

For a step h , write $E_h = e^{hD}$ for translation: $(E_h p)(x) = p(x + h)$. The complete binary signed difference operator is

$$\Delta_N^{\text{TM}}(b_1, \dots, b_N) := \prod_{j=1}^N (1 - E_{b_j}). \quad (69)$$

Expanding the product assigns a sign $(-1)^{|J|}$ to each subset J . With geometric steps, subset sums are binary integers and the signs are precisely Thue–Morse.

Theorem 5.3 (Uncentered Thue–Morse–Barnes collapse; NEW). *Let $\varepsilon_k = (-1)^{\text{wt}_2(k)}$. For every $N \geq 1$ and $n \geq 0$,*

$$\sum_{k=0}^{2^N-1} \varepsilon_k P_n^{[N]}\left(x + \frac{k}{2^N}\right) = (-1)^N 2^{-N(N+1)/2} n^N x^{n-N}, \quad (70)$$

where the right side is interpreted as zero when $n < N$.

Proof. Multiply the generating function (62) by the complete signed exponential sum:

$$\begin{aligned} \sum_{k=0}^{2^N-1} \varepsilon_k \frac{e^{(x+k/2^N)t}}{L_N(t)} &= \frac{e^{xt}}{L_N(t)} \prod_{j=1}^N (1 - e^{t/2^j}) \\ &= (-1)^N e^{xt} t^N \prod_{j=1}^N 2^{-j} \\ &= (-1)^N 2^{-N(N+1)/2} t^N e^{xt}. \end{aligned}$$

The coefficient of $t^n/n!$ is the stated monomial. □

Corollary 5.4 (Prouhet cancellation with a canonical correction). *For $n < N$, the complete Thue–Morse sum in (70) vanishes. For $n = N$, it is the constant*

$$(-1)^N N! 2^{-N(N+1)/2}. \quad (71)$$

For $n > N$, it is exactly the monomial response of an N th finite difference. Thus the finite Kabaya–Iri polynomial is a canonical polynomial on which the Prouhet cancellation defect is completely diagonalized.

The centered form is sign-cleaner because the support is arranged symmetrically.

Theorem 5.5 (Centered Thue–Morse–Rvachev collapse; New). *With $s_N = 1 - 2^{-N}$, for every $N \geq 1$ and $n \geq 0$,*

$$\sum_{k=0}^{2^N-1} \varepsilon_k A_n^{[N]} \left(x + s_N - \frac{k}{2^{N-1}} \right) = 2^{-N(N-1)/2} n^N x^{n-N}. \quad (72)$$

Proof. The binary subset sums of the period vector $(1, 1/2, \dots, 2^{1-N})$ are $k/2^{N-1}$. Hence

$$\begin{aligned} \sum_{k=0}^{2^N-1} \varepsilon_k \frac{e^{(x+s_N-k/2^{N-1})t}}{M_N(t)} \\ = \frac{e^{(x+s_N)t}}{M_N(t)} \prod_{j=1}^N (1 - e^{-a_j t}). \end{aligned}$$

Now

$$\prod_{j=1}^N (1 - e^{-a_j t}) = e^{-2s_N t} \prod_{j=1}^N (e^{a_j t} - 1),$$

while

$$M_N(t)^{-1} = e^{s_N t} \prod_{j=1}^N \frac{a_j t}{e^{a_j t} - 1}.$$

All exponentials cancel, leaving $e^{xt} t^N \prod_j a_j = 2^{-N(N-1)/2} t^N e^{xt}$. Coefficient comparison proves the theorem. $\square$

Example 5.6 (The first genuinely centered identity). For $N = 2$,

$$A_4^{[2]}(x) = x^4 - \frac{5}{8}x^2 + \frac{53}{1280}. \quad (73)$$

The four-point Thue–Morse difference is

$$\begin{aligned} A_4^{[2]}(x + 3/4) - A_4^{[2]}(x + 1/4) - A_4^{[2]}(x - 1/4) + A_4^{[2]}(x - 3/4) \\ = 6x^2. \end{aligned} \quad (74)$$

The constants and lower even powers disappear exactly, not asymptotically.

5.4 A q-binomial form of the difference operator

The Thue–Morse block polynomial

$$T_N(z) = \sum_{k=0}^{2^N-1} \varepsilon_k z^k = \prod_{r=0}^{N-1} (1 - z^{2^r}) \quad (75)$$

encodes the same operation as (69). The collapse theorems can therefore be read in three interchangeable languages:

$$\text{binary subset signs} \longleftrightarrow \text{Thue–Morse block polynomial} \longleftrightarrow \text{product of Barnes difference factors.} \quad (76)$$

This is a new bridge between the formula atlas and the finite convolution/Barnes hierarchy. It also suggests a higher-rank extension: replace the simple product of differences by the Pascal-weighted multiplicities of the repository’s rank- r Rvachev hierarchy, which should produce multiple Thue–Morse–Barnes operators with repeated geometric periods.

6 Exact finite-prefix expansions and q -Richardson recovery

6.1 Tail self-similarity is polynomial in the truncation scale

Let X', Y' be independent copies of X, Y , independent of the first N summands. Splitting the random series after level N gives the exact distributional identities

$$X \stackrel{d}{=} X_N + 2^{-N} X', \quad Y \stackrel{d}{=} Y_N + 2^{-N} Y'. \quad (77)$$

Hence

$$L(t) = L_N(t)L(2^{-N}t), \quad M(t) = M_N(t)M(2^{-N}t). \quad (78)$$

Write $\mu_r = \mathbb{E}X^r$ and $m_{2r} = \mathbb{E}Y^{2r}$.

Theorem 6.1 (Exact finite-prefix expansion; NEW). *For every $N, n \geq 0$,*

$$P_n^{[N]}(x) = \sum_{r=0}^n \binom{n}{r} \mu_r 2^{-Nr} P_{n-r}(x). \quad (79)$$

For the centered sequence,

$$A_n^{[N]}(x) = \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} m_{2r} 4^{-Nr} A_{n-2r}(x). \quad (80)$$

Thus $P_n^{[N]}(x)$ is exactly a polynomial of degree n in 2^{-N} , while $A_n^{[N]}(x)$ is exactly a polynomial of degree $\lfloor n/2 \rfloor$ in 4^{-N} .

Proof. From (78),

$$\begin{aligned} \frac{e^{xt}}{L_N(t)} &= L(2^{-N}t) \frac{e^{xt}}{L(t)}, \\ \frac{e^{xt}}{M_N(t)} &= M(2^{-N}t) \frac{e^{xt}}{M(t)}. \end{aligned}$$

Expand the first extra factor in moments and compare exponential generating coefficients. In the centered case all odd moments vanish, so only powers $2^{-2Nr} = 4^{-Nr}$ survive. $\square$

Remark 6.2. Earlier extrapolation results in the repository accelerate an infinite sinc product or extract periodic endpoint coefficients from asymptotic series. Here there is no asymptotic remainder at all. For each fixed polynomial degree, the dependence on the prefix depth is finite and algebraic. Richardson extrapolation becomes exact interpolation at the missing node 0.

6.2 Geometric Lagrange weights

Fix $0 < q < 1$ and an order $p \geq 0$. The Lagrange weights that evaluate a degree- p polynomial at zero from its values at $1, q, \dots, q^p$ are

$$w_{p,j}(q) = \prod_{\substack{0 \leq \ell \leq p \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell} \quad (0 \leq j \leq p). \quad (81)$$

They have the Gaussian-binomial form

$$w_{p,j}(q) = \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_j (q; q)_{p-j}} = \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_p} \begin{bmatrix} p \\ j \end{bmatrix}_q. \quad (82)$$

The interpolation moments are

$$\sum_{j=0}^p w_{p,j}(q) q^{jr} = \begin{cases} 1, & r = 0, \\ 0, & 1 \leq r \leq p. \end{cases} \quad (83)$$

Lean boundary for the weights. The source-level individual quotient is already represented by `geometricLagrangeWeight_eq_gaussianBinomial_div` in `GeometricQBinomialLagrange.lean`, over a field and under injectivity of $j \mapsto q^j$ on $\{0, \dots, p\}$. Its numerator deliberately uses `gaussianBinomialqp(p-j)`; the symmetric-index notation in (82) is the equivalent paper convention, not a newly added generic symmetry theorem. This closed-weight API predates the principal-specialization and generic-residual tranche described below. In the present real range $0 < q < 1$, its finite-node hypothesis is automatic.

Theorem 6.3 (Exact q -Richardson recovery; NEW). *For every $n, N \geq 0$,*

$$P_n(x) = \sum_{j=0}^n w_{n,j}(1/2) P_n^{[N+j]}(x). \quad (84)$$

Let $d = \lfloor n/2 \rfloor$. Then

$$A_n(x) = \sum_{j=0}^d w_{d,j}(1/4) A_n^{[N+j]}(x). \quad (85)$$

Both identities hold for every starting depth N , not merely in a limit.

Proof. In (79), set $z = 2^{-N}$. The right side is a polynomial in z of degree at most n , and its value at $z = 0$ is $P_n(x)$. The values at depths $N+j$ are the values at zq^j with $q = 1/2$. Lagrange interpolation at zero is scale-invariant, so the weights are $w_{n,j}(1/2)$. The centered proof is identical with $z = 4^{-N}$, degree d , and $q = 1/4$. $\square$

Richardson without an error term

For fixed degree n , the infinite Rvachev-Appell polynomial is a finite linear combination of the Barnes polynomials associated with any $d+1$ consecutive dyadic convolution prefixes. The outer weights are Gaussian q -binomial coefficients. The construction therefore joins four structures that the repository previously treated in separate contexts:

finite splines $\longleftrightarrow$ Barnes periods $\longleftrightarrow$ q -Lagrange interpolation $\longleftrightarrow$ the infinite Rvachev law.

6.3 Residual moments beyond the exact degree

For later perturbations it is useful to know every nonvanishing geometric moment of the weights. Let h_r be the complete homogeneous symmetric polynomial. For $r \geq p + 1$,

$$\sum_{j=0}^p w_{p,j}(q) q^{jr} = (-1)^p q^{\binom{p+1}{2}} h_{r-p-1}(1, q, \dots, q^p) = (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} r-1 \\ p \end{bmatrix}_q. \quad (86)$$

Indeed, the generating function of the left side is

$$\begin{aligned} \sum_{r \geq 0} \left(\sum_{j=0}^p w_{p,j}(q) q^{jr} \right) z^r &= \sum_{j=0}^p \frac{w_{p,j}(q)}{1 - q^j z} \\ &= 1 + (-1)^p q^{\binom{p+1}{2}} \frac{z^{p+1}}{\prod_{j=0}^p (1 - q^j z)}. \end{aligned}$$

This identity is the exact analogue of the transformed-moment formulas used in classical Richardson error analysis. Here it also quantifies what happens when a numerical or symbolic approximation replaces the exact prefix polynomial.

Post-snapshot Lean status. At source checkpoint 1e761ed77583e9870ceeaeb2c74ac824094f0f3f, `completeHomogeneousEval_geometric` proves

$$h_s(1, q, \dots, q^p) = \begin{bmatrix} s+p \\ p \end{bmatrix}_q$$

over every commutative semiring, without division or a distinct-node hypothesis. Its literal right-hand side is the recursive term `gaussianBinomialq(s+p)p`, which agrees with the displayed quotient in the present field setting. The Lagrange formula (86) is the surviving range of `sum_geometricLagrangeWeight_mul_pow_of_pos`; its field hypothesis and finite-node injectivity are exactly those needed to construct the weights. The generic theorem `sum_weight_mul_geometric_pow_of_pos` assumes only the low moments themselves and therefore permits repeated nodes over an arbitrary commutative ring.

There is also a denominator-free scaled exact-row form. If weights $a_0, \dots, a_p$ over a commutative ring satisfy

$$\sum_{j=0}^p a_j (cq^j)^d = 0^d \quad (0 \leq d \leq p),$$

then `sum_weight_mul_scaled_geometric_pow_succ_add` gives, for $s \geq 0$,

$$\sum_{j=0}^p a_j (cq^j)^{p+1+s} = c^{p+1+s} (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} p+s \\ p \end{bmatrix}_q.$$

Exactness is assumed directly after scaling, so c need not be a unit or even nonzero. These results discharge the finite principal-specialization and residual-moment algebra only. The prefix-polynomial identifications and [theorem 6.3](#) are not consequences of them and remain outside the Lean development at this checkpoint.

6.4 Worked centered recovery in degree six

For $A_6(0) = \alpha_6$, the exact prefix constants at depths 1, 2, 3, 4 are

$$-\frac{31}{1344}, \quad -\frac{2995}{86016}, \quad -\frac{210499}{5505024}, \quad -\frac{13784195}{352321536}. \quad (87)$$

Since $d = 3$, the $q = 1/4$ weights are

$$\left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835}\right). \quad (88)$$

Their exact combination is

$$\sum_{j=0}^3 w_{3,j}(1/4) A_6^{[1+j]}(0) = -\frac{2347}{59535} = A_6(0). \quad (89)$$

No limiting argument or numerical approximation enters.

6.5 Worked uncentered recovery in degree three

For $P_3(0) = -1/12$, the finite constants at depths 1, 2, 3, 4 are

$$0, \quad -\frac{3}{128}, \quad -\frac{49}{1024}, \quad -\frac{525}{8192}. \quad (90)$$

The $q = 1/2$ weights

$$\left(-\frac{1}{21}, \frac{2}{3}, -\frac{8}{3}, \frac{64}{21}\right) \quad (91)$$

recover the limit exactly:

$$\sum_{j=0}^3 w_{3,j}(1/2) P_3^{[1+j]}(0) = -\frac{1}{12}. \quad (92)$$

6.6 A finite algorithm for any fixed degree

Algorithm 6.4 (Exact Appell polynomial recovery). To compute $A_n(x)$ exactly:

1. Set $d = \lfloor n/2 \rfloor$ and choose any starting depth $N \geq 1$.
2. For $j = 0, \dots, d$, form the finite Barnes polynomial

$$A_n^{[N+j]}(x) = \mathfrak{B}_n^{(N+j)} \left(x + 1 - 2^{-(N+j)} \mid 1, \frac{1}{2}, \dots, 2^{1-N-j} \right).$$

3. Combine them with $w_{d,j}(1/4)$ from (82).

The result is exactly $A_n(x)$. The analogous uncentered algorithm uses $n + 1$ levels and $q = 1/2$.

Remark 6.5 (Why centering halves the interpolation order). The raw expansion contains every power 2^{-Nr} through $r = n$. Symmetry removes odd moments, so the centered expansion contains only 4^{-Nr} through $r = \lfloor n/2 \rfloor$. Centering thus halves both the polynomial degree in the tail parameter and the number of prefix levels required for exact recovery. This is the exact-polynomial counterpart of the conditioning improvement obtained by centering logarithmic sinc-product tails in the repository's acceleration reports.

7 Reciprocal-sinc poles and the asymptotics of Appell constants

7.1 The reciprocal moment generating function

Put

$$R(t) = \frac{1}{M(t)} = \sum_{n \geq 0} \alpha_n \frac{t^n}{n!}. \quad (93)$$

The function M is entire, but R is meromorphic. By (10), the zeros of M are the imaginary lifts of the integer zeros of the Rvachev Fourier product. For $n \neq 0$, set

$$T_n = 2\pi i n, \quad \mu(n) = \nu_2(n) + 1. \quad (94)$$

Then T_n is a zero of M and a pole of R of order $\mu(n)$.

7.2 Exact leading Laurent coefficient

Write $n = 2^v u$ with u odd and define

$$C_u = M(\pi i u) \neq 0. \quad (95)$$

The number C_u is a real sinc product:

$$C_u = \prod_{j \geq 1} \frac{\sin(\pi u / 2^j)}{\pi u / 2^j}. \quad (96)$$

Theorem 7.1 (Laurent coefficient from the odd core; NEW). *For every nonzero integer $n = 2^v u$ with u odd,*

$$\boxed{\lim_{t \rightarrow T_n} (t - T_n)^{\mu(n)} R(t) = -\frac{T_n^{\mu(n)}}{C_u}.} \quad (97)$$

Thus the pole order is controlled by the 2-adic valuation of n , while the leading amplitude is controlled only by the odd core u .

Proof. Write $t = 2^{v+1}s$ and $s_0 = \pi i u$. Iterating $M(2s) = (\sinh s/s)M(s)$ gives

$$M(2^{v+1}s) = M(s) \prod_{r=0}^v \frac{\sinh(2^r s)}{2^r s}. \quad (98)$$

At $s = s_0$, the factor $M(s_0) = C_u$ is nonzero. Every displayed hyperbolic-sine factor has a simple zero. Since u is odd,

$$\cos(\pi u) = -1, \quad \cos(2^r \pi u) = 1 \quad (r \geq 1).$$

Consequently

$$M(2^{v+1}s) \sim -C_u \left(\frac{s - s_0}{s_0} \right)^{v+1} = -C_u \left(\frac{t - T_n}{T_n} \right)^{\mu(n)}.$$

Taking reciprocals proves (97). □

Remark 7.2. The repository already identifies the zero multiplicity $\nu_2(n) + 1$ of the sinc product. The new step is to retain the leading Laurent amplitude and to propagate the full divisor into coefficient asymptotics for the polynomial deconvolution sequence.

7.3 A 2-adic hierarchy of exponentially small corrections

Let

$$r_N = [t^N]R(t) = \frac{\alpha_N}{N!}. \quad (99)$$

For a fixed positive integer n , the conjugate pair of poles $\pm T_n$ contributes to r_{2m} . The highest Laurent term in (97) gives

$$\frac{2(-1)^{m+\mu(n)+1}}{C_u} \binom{2m + \mu(n) - 1}{\mu(n) - 1} \frac{1}{(2\pi n)^{2m}}, \quad (100)$$

while the lower Laurent coefficients contribute the same exponential scale multiplied by polynomials in m of degree at most $\mu(n) - 2$.

Corollary 7.3 (Pole multiplicity becomes polynomial degree; NEW). *For each fixed $n = 2^v u$, the complete contribution of the pair $\pm 2\pi i n$ to $\alpha_{2m}/(2m)!$ has the form*

$$\frac{(-1)^m}{(2\pi n)^{2m}} \Pi_n(m), \quad (101)$$

where Π_n is a real polynomial of degree exactly $v = \nu_2(n)$. Its leading coefficient is

$$\frac{2(-1)^v}{C_u v!} (2m)^v. \quad (102)$$

Hence the binary valuation of the Fourier zero is visible as the polynomial degree of the corresponding exponentially small correction to the Appell constants.

7.4 The first three pole shells explicitly

Define

$$C = C_1 = M(\pi i) = 0.55377127588881009534421741634454888128 \dots, \quad (103)$$

$$C_3 = M(3\pi i) = -0.04620229913430671299257486221028792308 \dots, \quad (104)$$

$$\begin{aligned} \eta &= \left. \frac{d}{dx} \log M(ix) \right|_{x=\pi} = \sum_{j \geq 1} \left(2^{-j} \cot \frac{\pi}{2^j} - \frac{1}{\pi} \right) \\ &= -0.40861837118802483147907598511740221545 \dots, \end{aligned} \quad (105)$$

so

$$\begin{aligned} \pi\eta &= -1.2837124730461260445945438959726042474 \dots, \\ \frac{C}{C_3} &= -11.985794782182535885188878444711216907 \dots \end{aligned} \quad (106)$$

The nearest poles $\pm 2\pi i$ are simple. The poles $\pm 4\pi i$ are double. At $T = 4\pi i$, the principal part is

$$R(t) = \frac{16\pi^2/C}{(t-T)^2} + \frac{4\pi i(\pi\eta - 2)/C}{t-T} + \mathcal{O}(1). \quad (107)$$

The poles $\pm 6\pi i$ are simple with odd-core constant C_3 .

Theorem 7.4 (Three-shell asymptotic; NEW). *As $m \rightarrow \infty$,*

$$\boxed{\frac{\alpha_{2m}}{(2m)!} = \frac{2(-1)^m}{C(2\pi)^{2m}} \left[1 + (1 - 2m - \pi\eta)4^{-m} + \frac{C}{C_3}9^{-m} + \mathcal{O}(m^2 16^{-m}) \right]}. \quad (108)$$

In particular,

$$\alpha_{2m} \sim \frac{2(-1)^m (2m)!}{C(2\pi)^{2m}}. \quad (109)$$

Proof. A simple pole at T with residue a contributes $-aT^{-2m-1}$ to r_{2m} . Applying (97) at $\pm 2\pi i$ gives the leading term in (108).

For $T = 4\pi i$, use (107). The coefficient of t^{2m} in $D(t-T)^{-2} + E(t-T)^{-1}$ is

$$D(2m+1)T^{-2m-2} - ET^{-2m-1}.$$

Adding the conjugate pole and simplifying gives

$$\frac{2(-1)^m}{C(4\pi)^{2m}} (1 - 2m - \pi\eta),$$

which is the relative 4^{-m} correction. The simple poles at $\pm 6\pi i$ contribute $2(-1)^m/(C_3(6\pi)^{2m})$, giving the relative term $(C/C_3)9^{-m}$.

After subtracting these principal parts, the next singularities are the triple poles at $\pm 8\pi i$. Standard meromorphic coefficient estimates therefore give an absolute remainder of order $m^2(8\pi)^{-2m}$; division by the leading scale $(2\pi)^{-2m}$ yields $\mathcal{O}(m^2 16^{-m})$. $\square$

m	normalized exact value $\mathcal{R}_m$	three-shell approximation	difference
4	0.976481902429846	0.975843927464026	6.38×10^{-4}
6	0.997609767253799	0.997605306103035	4.46×10^{-6}
8	0.999790455959967	0.999790427625023	2.83×10^{-8}
10	0.999983101160970	0.999983100994111	1.67×10^{-10}
12	0.999998705566884	0.999998705565958	9.26×10^{-13}
20	0.99999999965697235338431	0.99999999965697235337828	6.03×10^{-22}

Table 2: Verification of [theorem 7.4](#). Here $\mathcal{R}_m = \frac{C(2\pi)^{2m}}{2(-1)^m} \frac{\alpha_{2m}}{(2m)!}$.

7.5 Why the correction polynomial is genuinely dyadic

The shell $n = 2$ is closer than the odd shell $n = 3$, but its pole is double. Its correction is not a constant multiple of 4^{-m} ; it carries the linear factor $1 - 2m - \pi\eta$. The next power-of-two shell $n = 4$ is triple and produces a quadratic polynomial times 16^{-m} . In general, the shell $n = 2^v$ produces a degree- v polynomial times 4^{-vm} . This is the coefficient-side analogue of the zero-counting identities in the spectral reports: 2-adic multiplicity does not disappear under inversion; it changes from a divisor multiplicity into a hierarchy of polynomially weighted exponentials.

7.6 A Mittag–Leffler frontier

The local theorem suggests a global expansion

$$R(t) = H(t) + \sum_{n \neq 0} \mathcal{P}_n(t), \quad (110)$$

where $\mathcal{P}_n$ is a suitably regularized principal part at $2\pi in$ and H is entire. A convergent, symmetry-preserving Mittag–Leffler decomposition would give exact rapidly convergent series for all α_{2m} , with each summand labeled by the odd core and 2-adic valuation of n . It would also expose whether the constants C_u satisfy additional cyclotomic or Thue–Morse product relations along arithmetic rays.

8 A Lambert- W threshold for the Appell coefficients

8.1 From pole asymptotics to factorial growth

By [\(109\)](#) and Stirling’s formula,

$$\begin{aligned} \log |\alpha_{2m}| &= \log \left(\frac{2(2m)!}{C(2\pi)^{2m}} \right) + o(1) \\ &= 2m \log \frac{m}{\pi e} + \frac{1}{2} \log(4\pi m) + \log \frac{2}{C} + o(1). \end{aligned} \quad (111)$$

Thus the constants grow faster than any fixed exponential in m , but slower than $(2m)!$ by the geometric factor $(2\pi)^{-2m}$.

For a continuous inversion, let $\mathfrak{m}(X)$ be the large positive real solution of

$$\frac{2\Gamma(2\mathfrak{m} + 1)}{C(2\pi)^{2\mathfrak{m}}} = X. \quad (112)$$

Let $H = \log X$ and

$$w = W\left(\frac{H}{2\pi e}\right), \quad m_0 = \frac{H}{2w}. \quad (113)$$

The identity $we^w = H/(2\pi e)$ implies

$$\log \frac{m_0}{\pi e} = w, \quad 2m_0 \log \frac{m_0}{\pi e} = H. \quad (114)$$

Theorem 8.1 (Lambert coefficient threshold; NEW). As $X \rightarrow \infty$,

$$\mathfrak{m}(X) = m_0 - \frac{\frac{1}{2} \log(4\pi m_0) + \log(2/C)}{2 \log(m_0/\pi)} + o(1), \quad m_0 = \frac{\log X}{2W(\log X/(2\pi e))}. \quad (115)$$

If

$$m_{\min}(X) = \min\{m \in \mathbb{N} : |\alpha_{2m}| \geq X\}, \quad (116)$$

then

$$m_{\min}(X) = \frac{\log X}{2W(\log X/(2\pi e))} + \mathcal{O}(1). \quad (117)$$

Proof. The leading equation obtained from (111) is $2m \log(m/(\pi e)) = H$, whose exact solution is m_0 by (114). Write the right side of (111) as $f(m) + g(m) + o(1)$, where

$$f(m) = 2m \log \frac{m}{\pi e}, \quad g(m) = \frac{1}{2} \log(4\pi m) + \log \frac{2}{C}.$$

Since

$$f'(m) = 2 \log \frac{m}{\pi},$$

one Newton correction gives

$$\mathfrak{m} = m_0 - \frac{g(m_0)}{f'(m_0)} + o(1),$$

which is (115). The exact coefficients differ from their leading envelope by a relative quantity tending to zero, and passage to the least integer changes the answer by at most a bounded amount, proving (117). $\square$

8.2 A second Lambert geometry in the same theory

The repository's endpoint asymptotics use the lower branch W_{-1} to invert $x = \lambda 2^{-\lambda}$ as $x \downarrow 0$. The present threshold uses the principal branch W to invert $H = 2m \log(m/(\pi e))$ as coefficient size grows. The two appearances have different origins:

$$\begin{aligned} \text{endpoint geometry} & : \text{ dyadic scale count balanced against a small spatial argument,} \\ \text{coefficient geometry} & : \text{ factorial growth balanced against a reciprocal-sinc pole radius.} \end{aligned} \quad (118)$$

The common feature is an unknown occurring both linearly and inside a logarithm. The branch is determined by the direction of inversion: a vanishing endpoint requires W_{-1} , while an increasing factorial threshold uses the principal branch.

What the Lambert law does and does not say

Equation (117) predicts where the absolute Appell constants first cross a large threshold. It is not an endpoint asymptotic for F , and it does not approximate F^{-1} . Its input is the nearest reciprocal-sinc pole, not the saddle point of a Laplace inversion. The connection is structural rather than an identification of the two asymptotic problems.

9 Synthesis, computational consequences, and formalization roadmap

9.1 The common dyadic algebra

The principal results can be compressed into the following dictionary.

Dyadic datum	Nonlinear inverse side	Polynomial deconvolution side
Finite scale N	The derivative jet of F at $a/2^N$ terminates after order N .	The random sum Y_N has N box factors and an N -period Barnes inverse.
Bell structure	Partial Bell polynomials recursively invert $F \circ G = \text{id}$.	Complete Bell polynomials invert the moment generating function M .
Bernoulli structure	Enters through exact dyadic values and the surrounding asymptotic machinery.	Gives every cumulant and the dyadic refinement recurrence of A_n .
Thue–Morse	Generates the signed global extension that makes the derivative tower terminate.	Expands the complete product of geometric difference operators and collapses Barnes polynomials.
q -binomial/Lagrange	Lagrange–Bürmann extracts inverse Taylor coefficients.	Gaussian q -binomial Lagrange weights recover the infinite polynomial exactly from prefix levels.
Infinite sinc product	Governs endpoint inversion indirectly through the Fabius probability law.	Its reciprocal poles directly govern the large-degree Appell constants.
Lambert W	Lower branch W_{-1} resolves the small-argument saddle scale.	Principal branch W resolves the factorial coefficient threshold.

9.2 Computational consequences

The results provide three exact computational modes.

Inverse derivatives at rational outputs. For a dyadic input $x_0 = a/2^N$, compute the finite rational jet $f_1, \dots, f_N$ by the repository’s exact evaluator. Formula (20) then gives any desired derivative of G at the rational output $F(x_0)$ without numerical inversion or numerical differentiation.

Appell polynomials without infinite products. For fixed n , construct $d + 1$ finite Barnes polynomials and apply (85). Every operation is rational. The depth N is arbitrary, which gives many independent exact certificates for the same polynomial.

Large degree without Bell explosion. For large m , use (108). It avoids expanding a complete Bell polynomial of order $2m$ and reveals the sign, scale, and first exponentially small corrections directly from the first three pole shells.

9.3 Suggested Lean module decomposition

The proofs are well suited to staged formalization.

Proposed module	Principal declarations
<code>DyadicInverseGerm.lean</code>	Define the finite dyadic jet polynomial; prove flat remainder; connect formal reversion to inverse derivatives.
<code>InverseBellLagrange.lean</code>	Lagrange–Bürmann coefficient formula, partial-Bell recurrence, rationality, and the Catalan quarter-point theorem.
<code>RvachevAppell.lean</code>	Define L, M, P_n, A_n as formal series; prove affine transport, Appell, mean-value, and Bernoulli refinement identities.
<code>RvachevLogisticDual.lean</code>	Logistic characteristic function, convergence of the dyadic series Z , moment representation, and sign alternation.
<code>DyadicBarnes.lean</code>	Normalized Barnes generating function and exact identification of $P_n^{[N]}, A_n^{[N]}$.
<code>ThueMorseBarnesDifference.lean</code>	Binary subset-product expansion and the two monomial collapse theorems.
<code>AppellRichardson.lean</code>	Reuse the compiled Gaussian q -binomial weights, principal specialization, and residual moments; formalize tail self-similarity, the finite-prefix polynomial expansion, and exact Appell/Rvachev recovery.
<code>ReciprocalSincPoles.lean</code>	Pole order, leading Laurent coefficient, first double-pole principal part, and shell transfer to coefficients.
<code>AppellLambertThreshold.lean</code>	Stirling envelope and the principal-branch Lambert inversion.

Table 4: A possible formalization plan.

The algebraic sections can be formalized almost entirely in formal power series and polynomials. The analytic burden first becomes substantial in the logistic infinite product, the meromorphic pole remainder, and the Stirling–Lambert threshold. A practical sequence is therefore:

1. formal algebra and exact rational examples;
2. finite probability and Barnes identities;
3. q -interpolation and Thue–Morse products;
4. analytic convergence and pole transfer.

9.4 Further frontier questions

1. Arithmetic of dyadic inverse shadows

Determine the denominators, Galois groups, and critical-value valuations of the jet polynomials $Q_{a/2^N}$. Does the signed global extension force systematic factorization of their discriminants? Is there a direct q -binomial formula for the coefficients of the inverse shadow, parallel to the exact $q = 1/2$ formulas for $F(a/2^N)$?

2. Infinite-order Barnes limit

Develop a natural category in which

$$\mathfrak{B}_n^{(N)}(x + s_N \mid 1, 1/2, \dots, 2^{1-N}) \longrightarrow A_n(x)$$

is an infinite-period Barnes polynomial. The exact recovery theorem suggests that, degree by degree, the limit is more rigid than ordinary convergence: it is a finite q -interpolation invariant.

3. Higher-rank Pascal–Rvachev Appell systems

For the rank- r products already developed in the repository, define the reciprocal Appell sequence, identify its finite prefixes with Barnes polynomials having Pascal multiplicities, and determine the corresponding digital difference operator. The expected pole order $\binom{\nu_2(n)+1}{r}$ should produce correction polynomials of degree one less than that number.

4. Exact Mittag–Leffler expansion

Construct a convergent principal-part expansion for $1/M(t)$. Such an expansion should give exact series for α_{2m} indexed by odd cores u and valuations v , and may connect the constants $C_u = M(\pi i u)$ to the repository’s arithmetic-ray cyclotomic norms.

5. Flat defects as resurgent data

At a dyadic point, F and G are determined to all algebraic orders by finite polynomials, yet the actual functions differ from those Taylor shadows by nonzero flat germs. Quantify these defects on each side of the point, identify their endpoint-rescaled copies, and ask whether their transseries are controlled by the same periodic Lambert phase that governs the endpoint asymptotic.

10 Conclusion

The Fabius–Rvachev corpus already displays a remarkable collection of exact structures. The new results in this report do not add another disconnected formula; they identify two inverse calculi that organize several existing themes at once.

On the nonlinear side, dyadic derivative termination means that the inverse Fabius function has an algebraic Taylor shadow at every exact dyadic output. Lagrange–Bürmann and partial Bell polynomials compute the full inverse jet from finitely many rational data. The quarter point produces Catalan numbers exactly. Nowhere analyticity survives as a nonzero flat defect between the true inverse and its convergent algebraic series.

On the linear side, the known Kabaya–Iri polynomials invert averaging by the Fabius law. Centering reveals a symmetric Rvachev–Appell sequence with rational Bernoulli cumulants, complete-Bell constants, and a dyadic logistic moment representation. Finite convolution prefixes are geometric Barnes–Bernoulli polynomials. Their complete Thue–Morse difference is a single monomial, and their dependence on prefix depth is an exact finite polynomial. Gaussian q -binomial Richardson weights therefore recover the infinite sequence exactly.

Finally, reciprocal-sinc poles transform the repository’s 2-adic Fourier zero divisor into a large-degree coefficient hierarchy. A pole of order $\nu_2(n) + 1$ yields a degree- $\nu_2(n)$ polynomial correction, the first three shells already giving high precision. Stirling inversion then opens a second Lambert- W geometry, distinct from but structurally parallel to the endpoint saddle.

The resulting picture joins inverse Fabius calculus, Appell and Barnes polynomials, Thue–Morse products, q -binomial interpolation, Bell and Bernoulli algebra, infinite sinc products, and Lambert W in one finite-to-infinite dyadic mechanism.

A Complete audited TeX inventory

This appendix lists the 57 distinct `.tex` paths included in the repository-wide overlap audit. Historical files are frozen versions and were treated as prior art even when their content has since been consolidated elsewhere.

A.1 Living top-level and canonical sources (6)

1. docs/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2. docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3. docs/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4. docs/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5. docs/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6. docs/non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex

A.2 Living Rvachev Fourier-decay drafts (12)

7. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
8. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
9. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/rvachev_up_fourier_decay.tex
16. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/rvachev_up_fourier_decay.tex

17. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay.tex
18. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay.tex

A.3 Bundled source-paper TeX (7)

19. docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
20. docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
21. docs/papers/arXiv-1502.06738v2/Lacunary_products.tex
22. docs/papers/arXiv-1702.05442v1/09-Function.tex
23. docs/papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24. docs/papers/arXiv-2211.13570v1/document.tex
25. docs/papers/ubiq15/ubiq15-corrected.tex

A.4 Frozen archive: early notes and standalone studies (5)

26. docs/archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
27. docs/archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
28. docs/archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
29. docs/archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
30. docs/archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex

A.5 Frozen archive: primary expositions and supplements (8)

31. docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
32. docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
33. docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
34. docs/archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
35. docs/archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
36. docs/archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
37. docs/archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex

38. docs/archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex

A.6 Frozen archive: q-calculus, walkthrough, and glossary (6)

39. docs/archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
40. docs/archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
41. docs/archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
42. docs/archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
43. docs/archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
44. docs/archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex

A.7 Frozen archive: research frontiers (13)

45. docs/archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
46. docs/archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
47. docs/archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
48. docs/archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
49. docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
50. docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
51. docs/archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
52. docs/archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
53. docs/archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
54. docs/archive/research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
55. docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex

56. docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
57. docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

B Symbolic and numerical verification notes

The proofs above are algebraic or analytic; computation was used only to detect normalization errors, check rational examples, and test the pole expansion before it was proved. The following compact SymPy fragment reproduces the centered cumulants, Appell constants, finite-prefix polynomials, and the degree-six exact recovery.

Listing 1: Exact symbolic checks for the centered Appell sequence.

```
import sympy as sp

t, x = sp.symbols("t x")

def kappa_y(n):
    if n == 1 or n % 2:
        return sp.Integer(0)
    r = n // 2
    return 2**(2*r - 1) * sp.bernoulli(2*r) / (r * (4**r - 1))

# Reciprocal-MGF constants alpha_n.
alpha = [sp.Integer(1)]
for n in range(1, 13):
    s = sum(sp.binomial(n-1, j-1) * kappa_y(j) * alpha[n-j]
            for j in range(1, n+1))
    alpha.append(sp.factor(-s))

assert alpha[2] == -sp.Rational(1, 9)
assert alpha[4] == sp.Rational(31, 675)
assert alpha[6] == -sp.Rational(2347, 59535)

# Finite centered inverse polynomials.
def M_N(N):
    ans = 1
    for j in range(1, N+1):
        b = sp.Rational(1, 2**j)
        ans *= sp.sinh(b*t)/(b*t)
    return ans

def A_finite(N, n):
    g = sp.series(sp.exp(x*t)/M_N(N), t, 0, n+1).removeO()
    return sp.expand(sp.factorial(n) * g.coeff(t, n))

A4N2 = A_finite(2, 4)
assert A4N2 == x**4 - sp.Rational(5,8)*x**2 + sp.Rational(53,1280)

eps = lambda k: (-1)**(bin(k).count("1"))
tm = sum(eps(k) * A4N2.subs(x, x + sp.Rational(3,4)-sp.Rational(k,2))
        for k in range(4))
assert sp.expand(tm) == 6*x**2

# q-Lagrange weights and exact degree-six recovery.
```

```

def weight(p, j, q):
    return sp.factor(sp.prod((-q**ell)/(q**j-q**ell)
                             for ell in range(p+1) if ell != j))

q = sp.Rational(1, 4)
w = [weight(3, j, q) for j in range(4)]
assert w == [-sp.Rational(1,2835), sp.Rational(4,135),
             -sp.Rational(64,135), sp.Rational(4096,2835)]

vals = [A_finite(N, 6).subs(x, 0) for N in range(1, 5)]
assert sp.factor(sum(w[j]*vals[j] for j in range(4))) == alpha[6]

```

The constants in [theorem 7.4](#) were computed directly from convergent products and sums:

Listing 2: High-precision evaluation of the pole constants.

```

import mpmath as mp
mp.mp.dps = 80

def M_ix(x, terms=500):
    p = mp.mpf(1)
    for j in range(1, terms+1):
        u = x/(2**j)
        p *= mp.sin(u)/u
    return p

C = M_ix(mp.pi)
C3 = M_ix(3*mp.pi)
eta = mp.nsum(lambda j: mp.cot(mp.pi/(2**j))/(2**j) - 1/mp.pi,
              [1, mp.inf])

print(C)
print(C3)
print(mp.pi*eta)
print(C/C3)

```

C A compact formula sheet

For convenient reuse, the main formulas are collected here.

$$\begin{aligned}
 Q_{a/2^N}(h) &= \sum_{k=1}^N \frac{2^{\binom{k+1}{2}} \widetilde{F}(2^k a/2^N)}{k!} h^k, \\
 G^{(m)}(F(a/2^N)) &= (m-1)! [u^{m-1}] (Q_{a/2^N}(u)/u)^{-m}, \\
 \frac{e^{xt}}{M(t)} &= \sum_{n \geq 0} A_n(x) \frac{t^n}{n!}, \quad A_n(x) = \mathbb{E}(x + iZ)^n, \\
 Z &= \sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j}, \quad \mathbb{E} e^{itZ} = M(t)^{-1}, \\
 A_n^{[N]}(x) &= \mathfrak{B}_n^{(N)} \left(x + 1 - 2^{-N} \mid 1, \frac{1}{2}, \dots, 2^{1-N} \right), \\
 \sum_{k < 2^N} \varepsilon_k A_n^{[N]} \left(x + 1 - 2^{-N} - \frac{k}{2^{N-1}} \right) &= 2^{-N(N-1)/2} n^N x^{n-N},
 \end{aligned}$$

$$\begin{aligned}
A_n^{[N]}(x) &= \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} m_{2r} 4^{-Nr} A_{n-2r}(x), \\
A_n(x) &= \sum_{j=0}^{\lfloor n/2 \rfloor} w_{\lfloor n/2 \rfloor, j}(1/4) A_n^{[N+j]}(x), \\
w_{p,j}(q) &= \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_j (q; q)_{p-j}}, \\
\lim_{t \rightarrow 2\pi i n} (t - 2\pi i n)^{\nu_2(n)+1} M(t)^{-1} &= -\frac{(2\pi i n)^{\nu_2(n)+1}}{M(\pi i n / 2^{\nu_2(n)})}, \\
\frac{\alpha_{2m}}{(2m)!} &= \frac{2(-1)^m}{C(2\pi)^{2m}} \left(1 + (1 - 2m - \pi\eta)4^{-m} \right) \\
&\quad + \frac{2(-1)^m}{C(2\pi)^{2m}} \left(\frac{C}{C_3} 9^{-m} + \mathcal{O}(m^2 16^{-m}) \right), \\
m_{\min}(X) &= \frac{\log X}{2W(\log X / (2\pi e))} + \mathcal{O}(1).
\end{aligned}$$

References

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